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4.1 Cell Biology

AQA GCSE Biology (8461) — Paper 1

4.1.1 Cell Structure

Cells are the smallest unit of life. All living things are made of cells. Most cells are so small they can only be viewed under a microscope. Biology divides all cells into two broad groups: **eukaryotic** (cells with a nucleus — plants, animals, fungi, protists) and **prokaryotic** (cells without a true nucleus — bacteria).

Animal and Plant (Eukaryotic) Cells

Organelle	Function	Animal	Plant
Nucleus	Contains DNA (chromosomes) which controls cell activities	✓	✓
Cytoplasm	Jelly-like fluid where chemical reactions occur	✓	✓
Cell membrane	Partially permeable — controls what enters/leaves	✓	✓
Mitochondria	Site of aerobic respiration (releases energy)	✓	✓
Ribosomes	Site of protein synthesis	✓	✓
Cell wall (cellulose)	Strong support, prevents bursting	✗	✓
Chloroplasts	Contain chlorophyll — photosynthesis	✗	✓

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Permanent vacuole	Filled with cell sap — support	✗	✓

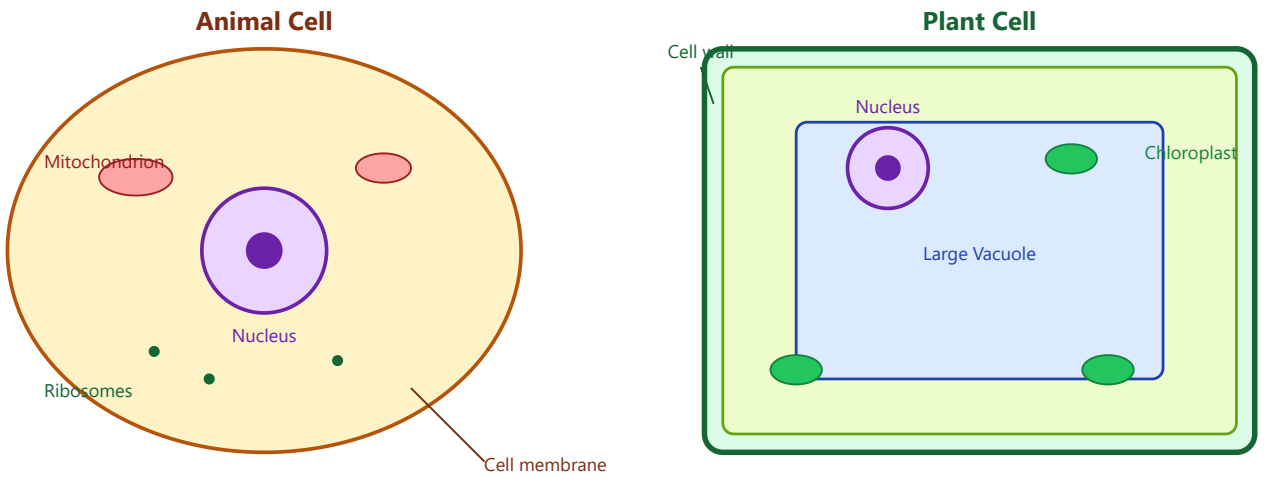

Fig 1: Animal vs plant cell — note that only plant cells have a cell wall, chloroplasts and a large permanent vacuole.

Prokaryotic Cells (Bacteria)

Prokaryotes are **much smaller** than eukaryotic cells (0.1–5 µm vs 10–100 µm) and have simpler internal organisation.

- **No nucleus** — their DNA is a single loop in the cytoplasm.
- **Plasmids** — small rings of extra DNA. Often carry antibiotic resistance genes.
- No membrane-bound organelles. Ribosomes are present.
- Surrounded by a **cell wall** (not made of cellulose).
- Some have a **flagellum** (tail for movement) or a **slime capsule** (protection).

> **Exam Tip:** When comparing cells, learn the three key features of eukaryotic cells: (1) a true nucleus, (2) genetic material enclosed inside it, (3) membrane-bound organelles. Prokaryotes lack all three.

Specialised Cells

During an organism's development, cells **differentiate** into specialised forms with specific functions. Most animal cells do this early; plant cells retain the ability to differentiate throughout life.

Cell	Function	Key adaptations
Sperm cell	Fertilise an egg	Tail (flagellum) for swimming; many mitochondria for energy; enzymes in acrosome to penetrate egg; haploid nucleus
Nerve cell (neurone)	Transmit electrical impulses	Long axon for rapid transmission; dendrites connect to many cells; myelin insulation
Muscle cell	Contract to produce movement	Packed with contractile proteins; many mitochondria
Root hair cell	Absorb water & mineral ions	Long projection — large surface area; thin wall; many mitochondria for active transport
Xylem cell	Transport water up the plant	Dead cells; lignin-strengthened walls; hollow tubes; no end walls
Phloem cell	Transport sugars around the plant	Living sieve tube cells; porous sieve plates; supported by companion cells

4.1.2 Microscopy

Light vs Electron Microscope

Feature	Light microscope	Electron microscope
Source	Light	Beam of electrons
Max magnification	~×1500	~×2 000 000
Max resolution	~200 nm	~0.2 nm

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Cost & use	Cheap, simple, can view living cells	Expensive, complex, cannot view living cells

Magnification = how much larger the image looks.
Resolution = how far apart two points can be and still be distinguished as separate. Electron microscopes have much better resolution — they can reveal ribosomes, inner mitochondrial membranes and detailed structure.

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Magnification equation:
Magnification = size of image ÷ size of real object
(keep units the same, e.g. both in µm)
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Standard Form and Units

- 1 millimetre (mm) = 1×10^{-3} m = 1000 µm
- 1 micrometre (µm) = 1×10^{-6} m = 1000 nm
- 1 nanometre (nm) = 1×10^{-9} m

Worked example: A cell has a real diameter of 20 µm. On a photograph it measures 80 mm. Calculate the magnification.

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Step 1 — same units: 80 mm = 80 000 µm.
Step 2 — divide: 80 000 ÷ 20 = ×4000.
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4.1.3 Cell Division by Mitosis

Mitosis is the division of a body cell to produce **two genetically identical diploid daughter cells**. It is used for growth, repair of damaged tissue and asexual reproduction.

Mitosis is part of the **cell cycle**, which has three main phases:

1. **Interphase** — cell grows, organelles duplicate, DNA replicates so each chromosome has two sister chromatids.
2. **Mitosis** — the nucleus divides: chromosomes line up, then sister chromatids are pulled apart to opposite poles.
3. **Cytokinesis** — cytoplasm and cell membrane divide giving two daughter cells.

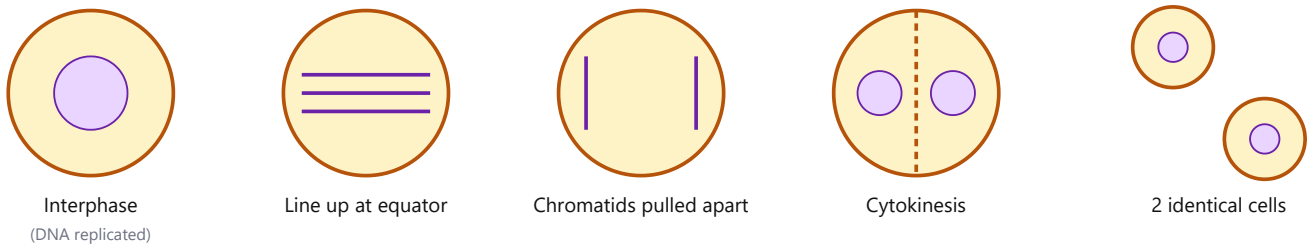

Fig 2: Main stages of mitosis. Each daughter cell receives a full copy of every chromosome.

Binary Fission in Bacteria

Bacteria reproduce **asexually** by binary fission: the DNA loop is copied, the cell elongates and splits in two. In ideal conditions a bacterium can divide every 20 minutes, producing millions within a day.

> **Bacterial growth:** number of bacteria = 2^n, where n = number of divisions. Example: 1 bacterium dividing every 20 min; after 2 h (120 min) → n = 6 divisions → 2^6 = 64 bacteria.

4.1.4 Stem Cells

> A **stem cell** is an undifferentiated cell that can divide by mitosis to form more of the same type of cell and can differentiate into specialised cells.

Source	Potential	Uses
Embryonic stem cells	Can become almost any cell type (pluripotent)	Potential treatment for many conditions

Adult stem cells (e.g. bone marrow)	Limited range of cell types	Treating blood disorders, leukaemia
Meristem tissue (plants)	Can become any plant cell	Cloning rare plants, disease-free crops

Therapeutic Cloning

An embryo is produced with the same genes as the patient, so stem cells obtained from it would not be rejected by the immune system. **Advantages**: personalised treatment, no rejection. **Disadvantages**: ethical concerns (destroying embryos), risk of tumours, high cost.

> **Exam Tip:** AQA expects balanced evaluation of stem cell use. Learn one benefit (e.g. paralysis, type 1 diabetes) and one objection (destruction of embryos / religious/ethical views) — you must mention both for full marks in a "discuss" question.

4.1.5 Transport In and Out of Cells

Diffusion

> **Diffusion** — the *net* movement of particles from a region of **higher** concentration to a region of **lower** concentration, down a concentration gradient. It is passive (no energy required).

Factors affecting diffusion rate: **steeper concentration gradient**, **higher temperature**, **larger surface area**, **thinner membrane**.

Osmosis

> **Osmosis** — the net movement of **water** across a **partially permeable membrane** from a more dilute solution (higher water concentration) to a more concentrated solution (lower water concentration).

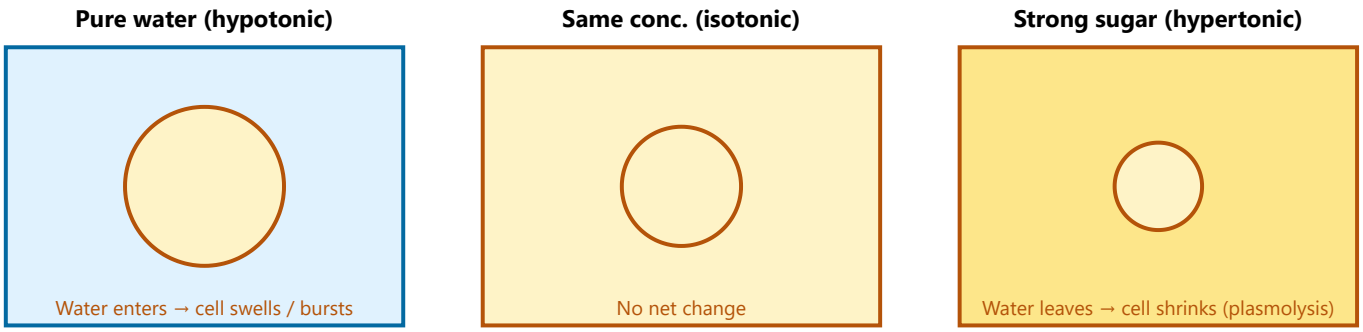

Fig 3: Three cases of osmosis — water movement depends on relative concentrations.

Active Transport

Active transport moves substances **against** their concentration gradient (low → high). It requires **energy from respiration (ATP)** and uses **carrier proteins** in the membrane.

Two examples:

- **Root hair cells** absorb mineral ions from the soil even when ion concentration is higher inside the cell than out.
- **Small intestine** absorbs glucose into the blood when blood glucose is already higher.

Process	Direction	Energy?	Example
Diffusion	High → low	No	O_2 into lung cells
Osmosis	High water → low water	No	Water into root hair cells
Active transport	Low → high	Yes (ATP)	Mineral ions into roots

Exam Tip: When giving examples of active transport, always name the substance, the type of cell and the direction. "Glucose absorbed from the small intestine into the blood, against the concentration gradient" earns full marks.

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Key Definitions — 4.1 Summary

- **Diffusion** — net movement of particles down a concentration gradient.
- **Osmosis** — diffusion of water across a partially permeable membrane.
- **Active transport** — movement against a gradient, requires ATP.
- **Mitosis** — cell division producing two identical diploid cells.
- **Differentiation** — how cells become specialised during development.
- **Stem cell** — undifferentiated cell capable of becoming different cell types.
- **Magnification** = image size ÷ real size.

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